Wk 4 Supervision Problems 2

Luca Bollini

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Sequences

1. Determine
$$\lim_{n\to\infty}(2^n+3^n)^{\frac{1}{n}}.$$

2. (i) Show that if (a_n) and (b_n) are convergent sequences with $a_n \to \alpha$, $b_n \to \beta$ then
$$\max\{a_n, b_n\} \to \max\{\alpha, \beta\}, \quad \min\{a_n, b_n\} \to \min\{\alpha, \beta\}.$$
(hint: think of a clever way to write $\max\{x, y\}$ *you could apply algebra of limits to)*

   (ii) Give an example of divergent sequences (a_n) and (b_n) for which $\max\{a_n, b_n\}$ and $\min\{a_n, b_n\}$ both converge. Show, however, that if (a_n), $\max\{a_n, b_n\}$ and $\min\{a_n, b_n\}$ all converge, then (b_n) is also convergent.

3. Let (a_n) and (b_n) be sequences of positive numbers such that, for $n \geq 1$,
$$a_{n+1} = \frac{1}{2}(a_n + b_n), \quad b_{n+1} = \sqrt{a_n b_n}$$

   (i) Show that for all $n \geq 2$, (a_n) is monotone decreasing and (b_n) is monotone increasing.

   (ii) Deduce that (a_n) and (b_n) have the same limit.

Bonus (Olympiad Style)

A $3 \times k$ board has every cell coloured black or white. Such a board is called:

- *sandwich-free* if there are no three adjacent cells in any row or column which form a "sandwich": a black cell surrounded by white cells or a white cell surrounded by black cells, i.e. ■□■ or □■□.
- *three-free* if there are no three adjacent cells in any row or column that are the same colour, i.e. ■■■ or □□□.
- *four-balanced* if every four adjacent cells in any row have the same number of black and white cells (two of each).

Let S be the number of sandwich-free 3×2026 boards, F be the number of three-free 3×2026 boards, and B be the number of four-balanced 3×2026 boards. Determine $S + B - F$.